Polynomial Division: Box Method – KEY

Name: ____________________ **Date:** __________ **Period:** _____

The Big Idea: Division is multiplication **backwards**!

Example 1: $(x^3 + 5x^2 + 8x + 4) \div (x + 1)$

$\times$	x^2	$4x$	4
x	x^3	$4x^2$	$4x$
$+1$	x^2	$4x$	4

Quotient: $x^2 + 4x + 4$ **Remainder:** **0**

Example 2: $(2x^4 + 3x^2 - 10) \div (x + 2)$ **Missing terms!*

Rewrite with placeholders: $2x^4 + 0x^3 + 3x^2 + 0x - 10$

$\times$	$2x^3$	$-4x^2$	$11x$	-22
x	$2x^4$	$-4x^3$	$11x^2$	$-22x$
$+2$	$4x^3$	$-8x^2$	$22x$	-44

Quotient: $2x^3 - 4x^2 + 11x - 22$ **Remainder:** **34**

Note: Box gives -44 but we need -10, so remainder $= -10 - (-44) = 34$

You Try: $(3x^3 + 10x^2 + x - 6) \div (x + 2)$

$\times$	$3x^2$	$4x$	-7
x	$3x^3$	$4x^2$	$-7x$
$+2$	$6x^2$	$8x$	-14

Quotient: $3x^2 + 4x - 7$ **Remainder:** **8**

Note: Box gives -14 **but we need** -6**, so remainder** $= -6 - (-14) = 8$